EXTREMAL GRAPH THEORY

Erdős Problem 915

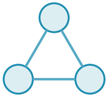
Maximizing Edge Density Subject to Connectivity Constraints

Louis Vandenbruwaene

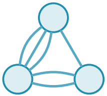
Supervisor: Prof. Stijn Cambie   Promotor: Prof. Jan Goedgebeur

KU Leuven, Master of Mathematics

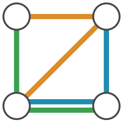
The eight models



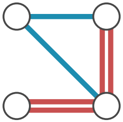
simple graph
one edge per pair



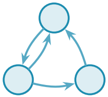
multigraph
parallel edges $\mu(u,v) \geq 1$



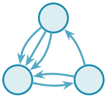
hypergraph
hyperedges span $r \geq 3$



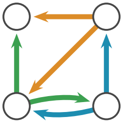
multihypergraph
a hyperedge repeated



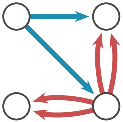
simple digraph
one arc per ordered pair



directed multigraph
parallel and opposite arcs

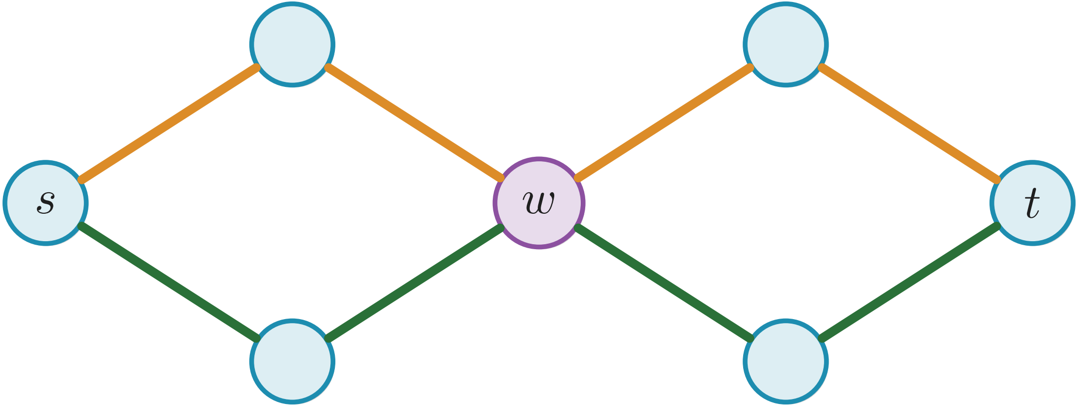


directed hypergraph
two share an edge



directed multihypergraph
a hyperedge repeated

The separation axis

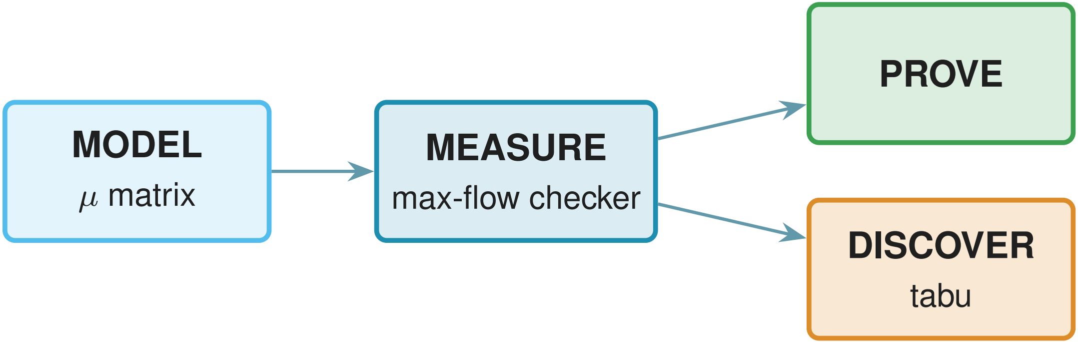


$\lambda(s,t) = 2$ and $\kappa(s,t) = 1$.

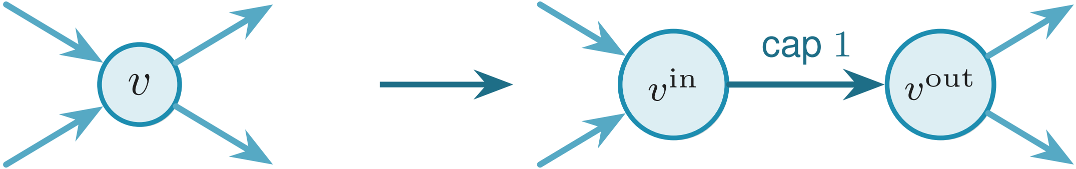
Notation for the extremal functions

Model	edge-disjoint routes	internally vertex-disjoint routes
Simple graph	$\ell_m(n)$	$k_m(n)$
Multigraph	$L_m(n)$	$K_m(n)$
r-uniform simple hypergraph	$\ell_m^{(r)}(n)$	$k_m^{(r)}(n)$
r-uniform multihypergraph	$L_m^{(r)}(n)$	$K_m^{(r)}(n)$
Simple digraph	$\ell_m^{\mathrm{dir}}(n)$	$k_m^{\mathrm{dir}}(n)$
Directed multigraph	$L_m^{\mathrm{dir}}(n)$	$K_m^{\mathrm{dir}}(n)$
Directed r-uniform simple hypergraph	$\ell_m^{\mathrm{dir}\,(r)}(n)$	$k_m^{\mathrm{dir}\,(r)}(n)$
Directed r-uniform multihypergraph	$L_m^{\mathrm{dir}\,(r)}(n)$	$K_m^{\mathrm{dir}\,(r)}(n)$

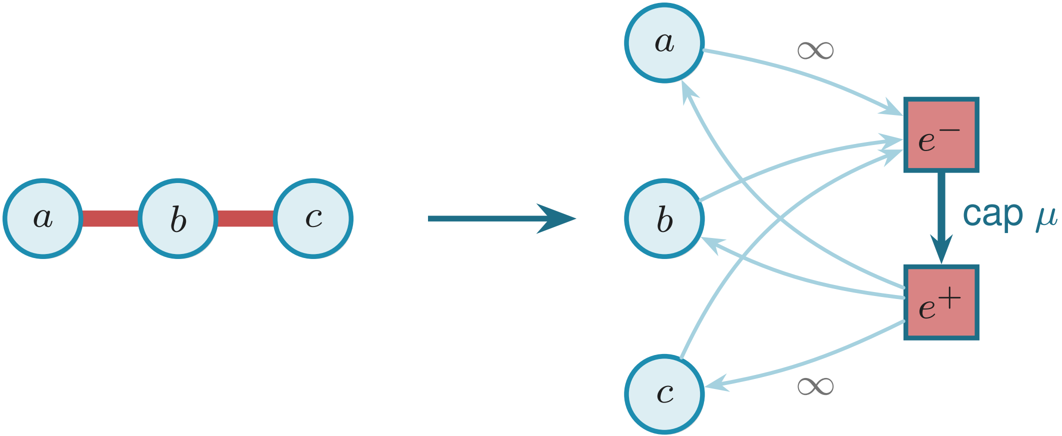
The architecture



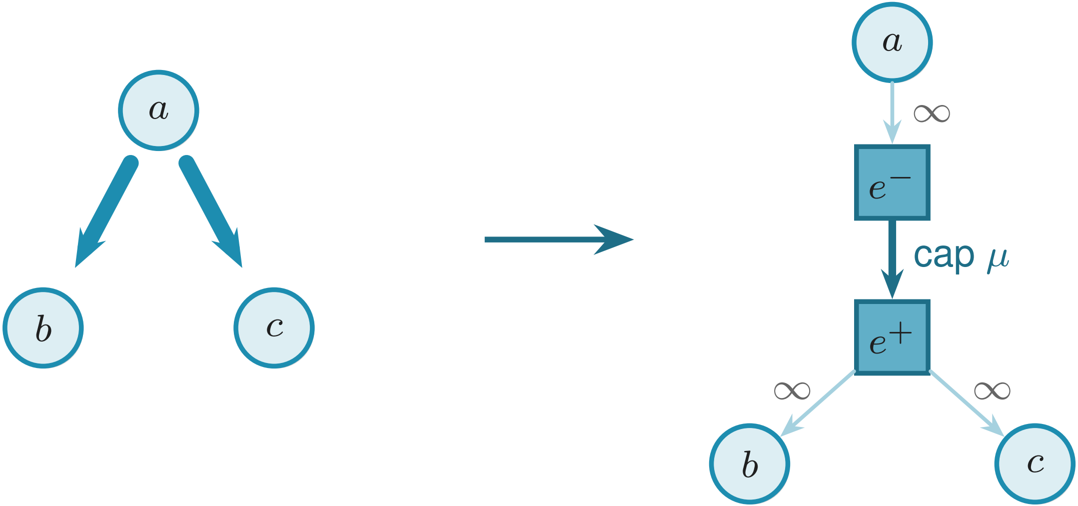
Transformations



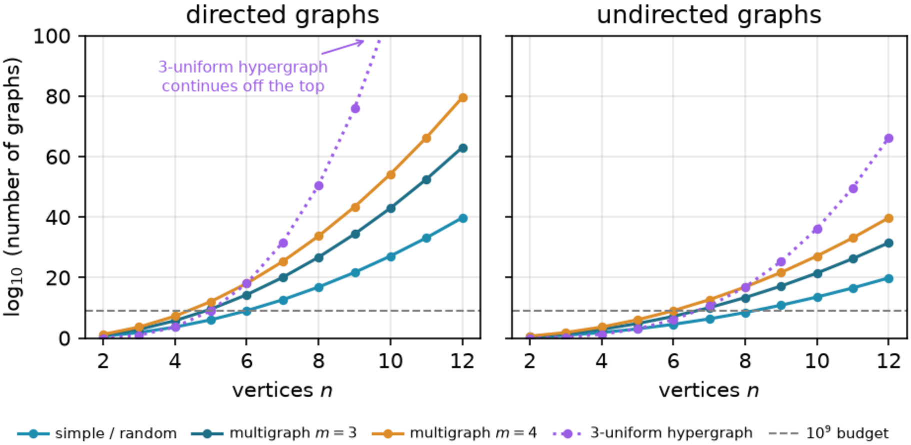
Transformations



Transformations



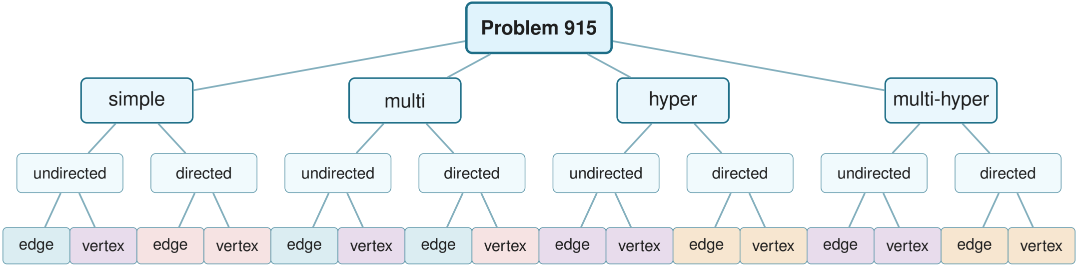
Candidate counts



Generating search: a benchmark

n	$\ell_2^{\mathrm{dir}}(n)$	blind (s)	pruned (s)	generated (s)
3	4	0.05	0.003	0.009
4	6	6.3	0.008	0.008
5	8	–	0.22	0.039
6	10	–	14.5	0.73
7	12	–	1156.6	17.0

Status of the sixteen variants



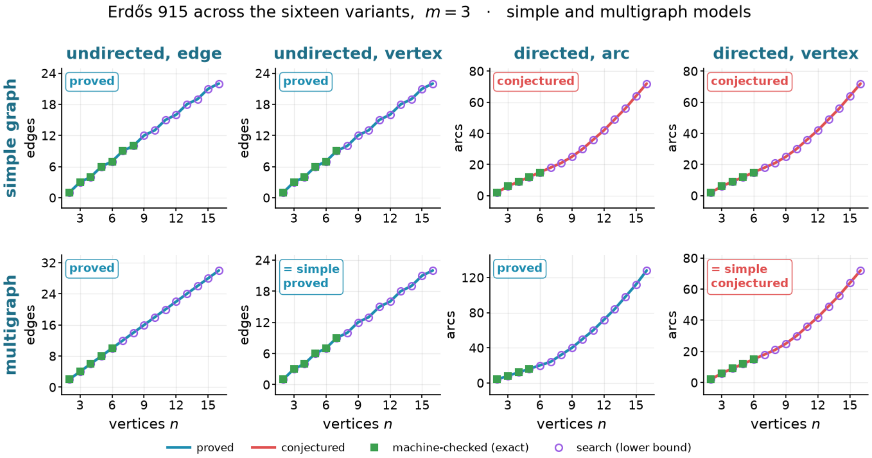
closed form proved for every n and m

leading constant proved, exact value open

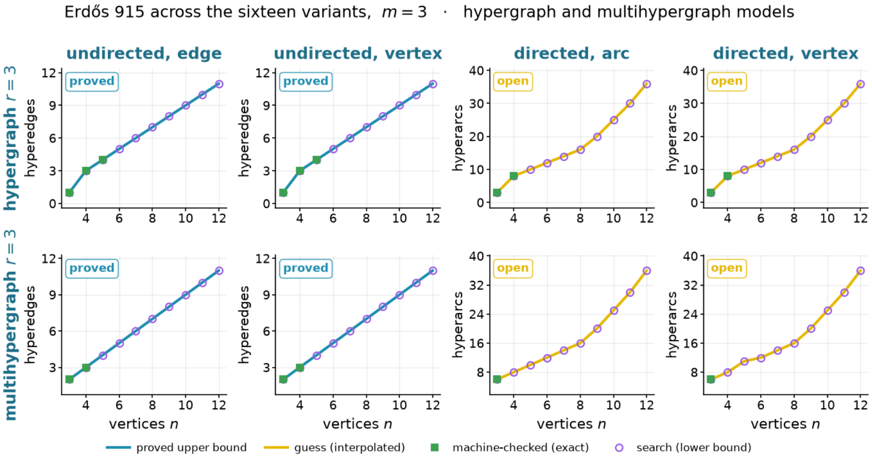
conjectured, matching certified small n

exact on stated parameter ranges, open outside them

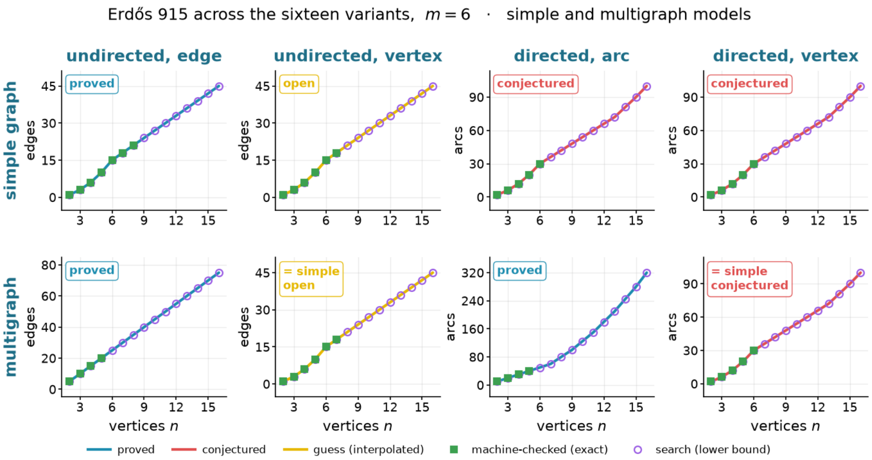
Eight graph variants at $m = 3$



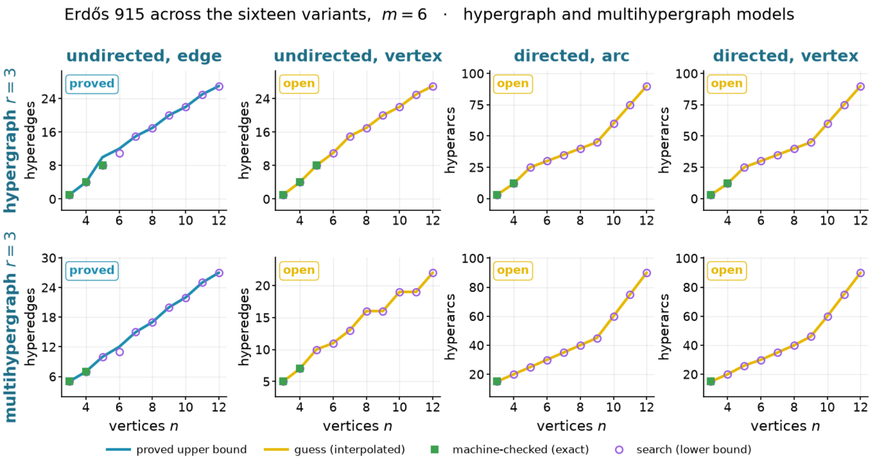
Eight hypergraph variants at $m = 3$



Eight graph variants at $m = 6$



Eight hypergraph variants at $m = 6$



The sixteen variants, summarised

Model	Edge or arc-disjoint	Vertex-disjoint
Undirected		
simple graph	$\lfloor m(n-1)/2 \rfloor$, proved	the edge value for $m \le 4$, and $\lfloor 8n/3 \rfloor - 4$ at $m = 5$; open for $m \ge 6$
multigraph	$(m-1)(n-1)$, proved	the simple value
r-hypergraph	$\le \lfloor (m-1)(n-1)/(r-1) \rfloor$, with equality in the range below	$\lfloor (n-1)/(r-1) \rfloor$ at $m = 2$ and $\le \lfloor 2(n-1)/(r-1) \rfloor$ at $m = 3$; open for $m \ge 4$
r-multi-hyper	the same bound, with equality when $(r-1) \mid (n-1)$	the same bounds, repeats closing cases the simple model misses at $m = 3$
Directed		
simple digraph	$M(n)$ at $m = 2$, proved; $n^2/4 + \Theta_m(n)$ for $m \ge 3$, exact value conjectured	$M(n)$ at $m = 2$, proved; $n^2/4 + \Theta_m(n)$ for $m \ge 3$, equality with the arc value open
multigraph	$(m-1)M(n)$, proved	the simple digraph value
r-hypergraph	$(1 + o(1))(m-1)n^2/[4(r-1)]$ for fixed r, m, leading term proved	the same leading term for fixed r, m
r-multi-hyper	the same leading term, the bound proved with repeats allowed	the same leading term for fixed r, m

Exact values, $m = 3$

Model	Variant	$n=2$	$n=3$	$n=4$	$n=5$	$n=6$	$n=7$	$n=8$
simple graph	undirected, edge ($\ell_m(n)$), proved	1	3	4	6	7	9	10
	undirected, vertex ($k_m(n)$), proved	1	3	4	6	7	9	10
	directed, arc ($\ell_m^{\mathrm{dir}}(n)$), conjectured	2	6	9	12	15	$\geq$18	$\geq$21
	directed, vertex ($k_m^{\mathrm{dir}}(n)$), conjectured	2	6	9	12	15	$\geq$18	$\geq$21
multigraph	undirected, edge ($L_m(n)$), proved	2	4	6	8	10	12	14
	undirected, vertex ($k_m(n)$), = simple, proved	1	3	4	6	7	9	10
	directed, arc ($L_m^{\mathrm{dir}}(n)$), proved	4	8	12	16	20	24	32
	directed, vertex ($k_m^{\mathrm{dir}}(n)$), = simple, conjectured	2	6	9	12	15	$\geq$18	$\geq$21
hypergraph $r = 3$	undirected, edge, proved		1	3	4	5	6	7
	undirected, vertex, proved		1	3	4	5	6	7
	directed, arc, open		3	8	$\geq$10	$\geq$12	$\geq$14	$\geq$16
	directed, vertex, open		3	8	$\geq$10	$\geq$12	$\geq$14	$\geq$16
multihypergraph $r = 3$	undirected, edge, proved		2	3	4	5	6	7
	undirected, vertex, proved		2	3	4	5	6	7
	directed, arc, open		6	$\geq$8	$\geq$10	$\geq$12	$\geq$14	$\geq$16
	directed, vertex, open		6	$\geq$8	$\geq$10	$\geq$12	$\geq$14	$\geq$16

Exact values, $m = 6$

Model	Variant	$n=2$	$n=3$	$n=4$	$n=5$	$n=6$	$n=7$	$n=8$
simple graph	undirected, edge ($\ell_m(n)$), proved	1	3	6	10	15	18	21
	undirected, vertex ($k_m(n)$), open	1	3	6	10	15	18	$\geq$21
	directed, arc ($\ell_m^{\mathrm{dir}}(n)$), conjectured	2	6	12	20	30	$\geq$36	$\geq$42
	directed, vertex ($k_m^{\mathrm{dir}}(n)$), conjectured	2	6	12	20	30	$\geq$36	$\geq$42
multigraph	undirected, edge ($L_m(n)$), proved	5	10	15	20	25	30	35
	undirected, vertex ($k_m(n)$), = simple, open	1	3	6	10	15	18	$\geq$21
	directed, arc ($L_m^{\mathrm{dir}}(n)$), proved	10	20	30	40	50	60	80
	directed, vertex ($k_m^{\mathrm{dir}}(n)$), = simple, conjectured	2	6	12	20	30	$\geq$36	$\geq$42
hypergraph $r = 3$	undirected, edge, proved		1	4	8	$\leq$12	15	17
	undirected, vertex, open		1	4	8	$\geq$11	$\geq$15	$\geq$17
	directed, arc, open		3	12	$\geq$25	$\geq$30	$\geq$35	$\geq$40
	directed, vertex, open		3	12	$\geq$25	$\geq$30	$\geq$35	$\geq$40
multihypergraph $r = 3$	undirected, edge, proved		5	7	10	$\leq$12	15	17
	undirected, vertex, open		5	7	$\geq$10	$\geq$11	$\geq$13	$\geq$16
	directed, arc, open		15	$\geq$20	$\geq$25	$\geq$30	$\geq$35	$\geq$40
	directed, vertex, open		15	$\geq$20	$\geq$26	$\geq$30	$\geq$35	$\geq$40

What remains open

Problem	Known here	Remaining target
Directed simple arc, $m \geq 3$	$\ell_m^{\mathrm{dir}}(n) = n^2/4 + \Theta_m(n)$	Prove the decomposition, or otherwise determine the linear coefficient.
Directed simple vertex, $m \geq 3$	$k_m^{\mathrm{dir}}(n) = n^2/4 + \Theta_m(n)$	Decide whether its exact value equals the arc value, given that an arc upper bound does not transfer.
Directed multigraph classification	Exact value for all n, m and quadratic-branch uniqueness for $n \geq \max(8, 2m)$	Classify the range $n < 2m$ and the remaining linear-branch extremisers.
Undirected simple vertex, $m \geq 6$	Block formulation and $m/2 < c_m \leq 2(m-1)$, where $c_m = \lim_n k_m(n)/(n-1)$	Determine c_m, whose likely next structure is triconnected rather than merely blockwise.
Multigraph vertex, parallel copies distinct	Block-knapsack formulation and order $\Theta(m^2 n)$ as $m \to \infty$ with $n/m \to \infty$, with the per-vertex constant between $3 - 2\sqrt{2}$ and 2	Determine the asymptotic constant and the optimal 2-connected blocks.
Hypergraph vertex, $m \geq 4$	Exact at $m = 2$ and, in the stated range, $m = 3$	Locate the first failure for $r \geq 3$ and replace the rank-two decomposition used at $m = 3$.
Directed hypergraphs	For fixed r, m, leading term $(m-1)n^2/(4(r-1))$ for all orientations and both separations	Determine finite exact values and whether forward and general agree once $n > r$.

AI: curse or blessing?

Pros

- LaTeX
- length & complexity of generated text

Cons

- control
- length & complexity of generated text

It's a tool.

short leash $+$ AI proofs $=$ argumented conjecture [AI]